ASSIGNMENT - 8

① (a) (i) Light-cone coordinates, (x^+ and x^-) are named ~~defined~~ as: so because they simplify the representation of light-like intervals.

(ii) In spacetime events separated by light signals, they satisfy the condition $ds^2 = 0$, where ds^2 is the spacetime interval.

(iii) For a light signal travelling in the x^1 direction, this implies $x^2 = \pm x^1$, which directly corresponds to $x^+ = 0$ or $x^- = 0$. This properly places ~~like~~ light like events along the co-ordinate axes in the $x^+ - x^-$ plane, making the light cone structure explicit.

(b) (i) The spacetime interval in the usual coordinates (x^0, x^1, x^2, x^3):

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$

In terms of light cone coordinates x^+ and x^-

$$(ds)^2 \cancel{(dx^0)^2} = \left(\frac{dx^+ + dx^-}{2}\right)^2 - \left(\frac{dx^+ - dx^-}{2}\right)^2 - (dx^2)^2 - (dx^3)^2$$

~~Expa~~ Expanding and simplifying :

$$ds^2 = (dx^+)(dx^-) - (dx^2)^2 - (dx^3)^2$$

Therefore, in light cone coordinates, the spacetime interval :

$$ds^2 = (dx^+)(dx^-) - (dx^2)^2 - (dx^3)^2$$

(ii) The metric tensor η in the light cone coordinates can be written in matrix form, taking into account that :

$$ds^2 = (dx^+)(dx^-) - (dx^2)^2 - (dx^3)^2$$

Thus the metric in the (x^+, x^-, x^2, x^3) basis is:

$$\eta_{\mu\nu} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

(c) For ~~S.L.T.~~ S.L.T. :-

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$$

$$\begin{bmatrix} x^{+\prime} \\ x^{-\prime} \\ x^{2\prime} \\ x^{3\prime} \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x^{+} \\ x^{-} \\ x^{2} \\ x^{3} \end{bmatrix}$$

$$x^{+\prime} = \gamma x^{+} - \beta\gamma x^{-}$$

$$= \gamma(x^0 + x^1) - \gamma\beta(x^0 - x^1)$$

$$= \gamma x^0(1-\beta) + \gamma x^1(1+\beta)$$

$$x^{-\prime} = \gamma x^{-} - \gamma\beta x^{+}$$

$$= \gamma(x^0 - x^1) - \gamma\beta(x^0 - x^1)$$

$$= \gamma x^0(1-\beta) - x^1\gamma(1+\beta)$$

$$x^{2\prime} = x^2 \quad ; \quad x^{3\prime} = x^3$$

(d) General 4-vector in light cone coordinates:

$$A^{\mu} = (A^{+}, A^{-}, A^{2}, A^{3})$$

Corresponding 4-vector: $A_{\mu} = (A_{+}, A_{-}, A_{2}, A_{3})$

For old coordinates, $\tilde{x}\cdot\tilde{x}$ should be invariant

$$\tilde{x}\cdot\tilde{x} = \tilde{x}^T \eta \tilde{x}$$

For now, light cone coordinates, $\tilde{x}\cdot\tilde{x}$ should be invariant. Let us call a new metric ϕ.

$$\tilde{x}\cdot\tilde{x} = \tilde{x}^T \phi \tilde{x}$$

$$= (x^{+} \; x^{-} \; x^{2} \; x^{3}) \, \phi \begin{bmatrix} x^{+} \\ x^{-} \\ x^{2} \\ x^{3} \end{bmatrix}$$

We know

$$\tilde{x}\cdot\tilde{x} = x^{+}x^{-} - (x^2)^2 - (x^3)^2$$

We need to form ϕ to satisfy this condit^n:

$$\tilde{x}\cdot\tilde{x} = \frac{x^{+}x^{-}}{2} + \frac{x^{+}x^{-}}{2} - (x^2)^2 - (x^3)^2$$

Now,

$$\phi \begin{bmatrix} x^+ \\ x^- \\ x^2 \\ x^3 \end{bmatrix} = \begin{bmatrix} \frac{x^-}{2} \\ \frac{x^+}{3} \\ -x^2 \\ -x^3 \end{bmatrix}$$

$$\Rightarrow \begin{pmatrix} x^+ & x^- & x^2 & x^3 \end{pmatrix} \begin{bmatrix} \frac{x^-}{2} \\ \frac{x^+}{2} \\ -x^2 \\ -x^3 \end{bmatrix}$$

$$= \cancel{\frac{x^+}{2}} \; \frac{x^+x^-}{2} + \frac{x^+x^-}{2} - (x^2)^2 - (x^3)^2$$

We construct ϕ accordingly :

$$\phi = \begin{bmatrix} 0 & 1/2 & 0 & 0 \\ 1/2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

② In relativistic world :

$$T = \int_{\vec{v}=\vec{0}}^{\vec{v}=\vec{u}} \frac{d}{dt}\left(m\gamma(v)\vec{v}\right) d\vec{r}$$

$$= \int_{\vec{v}=\vec{0}}^{\vec{v}=\vec{u}} d\left(m\gamma(v)\vec{v}\right) \vec{v}$$

$$\Rightarrow T = \int_0^u d\left(m\gamma(v)\, v\right) v$$

$$= \int_0^u mv \; d\left(\gamma(v)\, v\right)$$

$$\therefore d\left(\gamma(v)\, v\right) = -\frac{1}{2} \cdot v \frac{1}{\left(1-\beta_v^2\right)^{3/2}} - \frac{2v}{c^2}\, dv + \gamma(v)\, dv$$

$$= \beta_v^2 \gamma^3(v)\, dv + \gamma(v)\, dv$$

$$= \left[\beta_v^2 \cancel{\gamma^3(v)} + \gamma^{-2}(v)\right] \gamma^3(v)\, dv$$

$$= \gamma^3(v)\left[\beta_v^2 + 1 - \beta_v^2\right] dv$$

$$\therefore \boxed{d(\gamma(v)\, v) = \gamma^3(v)\, dv}$$

$$\therefore T = \int_0^u mv\gamma^3(v)\, dv = m\int_0^u \frac{v}{(1-\beta_v^2)^{3/2}}\, dv$$

$$1 - \frac{v^2}{c^2} = dx \qquad \text{or} \qquad v\, dv = -\frac{c^2}{2}\, dx$$

$$\cancel{\therefore T = -mc\int_0^{1-\beta_v^2}} \qquad \therefore T = -\frac{mc^2}{2}\int_1^{(1-\beta_v^2)} (1-\beta_v^2)^{-3/2}\, dx$$

$$= -\frac{mc^2}{2}\int_1^{(1-\beta_v^2)} x^{-3/2}\, dx$$

$$\text{or} \quad T = -\frac{mc^2}{2}\left[x^{-1/2}\right]_1^{1-\beta_u^2}$$

$$\text{or} \quad T = m\left[\frac{1}{\sqrt{1-\beta_u^2}} - 1\right] c^2$$

$$\boxed{T = m(\gamma(u) - 1)\, c^2} \longrightarrow (a)$$

(b) If $u << c$; then $\gamma(u) = \left(1 - \frac{u^2}{c^2}\right)^{-1/2} = 1 + \frac{u^2}{2c^2}$

$$\therefore T = m\left(1 + \frac{u^2}{2c^2} - 1\right) c^2$$

$$\boxed{T = \frac{1}{2} mu^2} \longrightarrow \text{reduces to Newtonian Formula}$$

$$\therefore T = mc^2\left[\left(1 - \frac{u^2}{c^2}\right)^{-1/2} - 1\right]$$

$$\simeq mc^2\left[\cancel{1} - \frac{1}{2}\left(-\frac{u^2}{c^2}\right) + \frac{3}{8}\left(-\frac{u^2}{c^2}\right) - \cancel{1}\right]$$

$$= mc^2\left(\frac{u^2}{2c^2} + \frac{3}{8}\frac{u^4}{c^4}\right)$$

$$\therefore \boxed{T \simeq mc^2\left(\frac{u^2}{2c^2} + \frac{3u^4}{8c^4}\right)} \text{ — } (b)$$

$\therefore u_1 = 0.01c$

$u_2 = 0.1c$

$u_3 = 0.5c$

$\beta_{u_1} = 0.01$

$\beta_{u_2} = 0.1$

$\beta_{u_3} = 0.5$

$\therefore \gamma(u_1) = (1-\beta_{u_1}^2)^{-1/2} = (1-0.0001)^{-1/2}$

$= 1.000050$

$\therefore \gamma(u_2) = (1-\beta_{u_2}^2)^{-1/2}$

$= (1-0.01)^{-1/2} = 1.0050$

$\therefore \gamma(u_3) = (1-\beta_{u_3}^2)^{-1/2}$

$= (1-0.25)^{-1/2}$

$= 1.1547$

$\therefore T_R = m(\gamma(u)-1)c^2$

$\therefore T_N = \frac{1}{2}mu^2$

For u_1: $T_R = 0.00005\ mc^2$

$T_N = 0.00005\ mc^2$

Error $= 0\%$

For u_2: $T_R = 0.0028\ mc^2$

$T_N = 0.002\ mc^2$

$\therefore$ Error $= 20\%$

For u_3: $T_R = \cancel{0.00}\ 0.1547\ mc^2$

$T_N = 0.125\ mc^2$

Error $= 19.198\ \%$